

Lab 5

实验目标

在数码管上，以0.5Hz滚动显示一个日期，日期可以自己随意选择。

实验代码设计

1. 亮起来

代码设计

```
module Hex7seg(  
    input clk,  
    input reset,  
    input [31:0] hex,  
    output [6:0] a_to_g0,  
    output [6:0] a_to_g1,  
    output [7:0] an  
);  
  
// 两部分数码管分别控制显示  
Hex7segIndex hex7segindex0(  
    .hex(hex[15:0]),  
    .clk(clk),  
    .reset(reset),  
    .a_to_g(a_to_g0),  
    .an(an[3:0])  
);  
  
Hex7segIndex hex7segindex1(  
    .hex(hex[31:16]),  
    .clk(clk),  
    .reset(reset),  
    .a_to_g(a_to_g1),  
    .an(an[7:4])  
);
```

```

    );
endmodule

module Hex7segIndex(
    input [15:0] hex,
    input clk,
    input reset,
    output reg [6:0] a_to_g,
    output reg [3:0] an
);

    wire [1:0] s;
    reg [3:0] digit;
    reg [19:0] clkdiv;
    assign s = clkdiv[19:18];

    always @ (*) begin // 数字译码
        case (s)
            2'b00: digit = hex[3:0];
            2'b01: digit = hex[7:4];
            2'b10: digit = hex[11:8];
            2'b11: digit = hex[15:12];
            default: digit = hex[3:0];
        endcase
    end

    always @(*) begin // 显示数字
        case (digit)
            4'b0000: a_to_g = 7'b1111110;
            4'b0001: a_to_g = 7'b0110000;
            4'b0010: a_to_g = 7'b1101101;
            4'b0011: a_to_g = 7'b1111001;
            4'b0100: a_to_g = 7'b0110011;
            4'b0101: a_to_g = 7'b1011011;
            4'b0110: a_to_g = 7'b1011111;
            4'b0111: a_to_g = 7'b1110000;
            4'b1000: a_to_g = 7'b1111111;
            4'b1001: a_to_g = 7'b1111011;
            4'b1010: a_to_g = 7'b1110111;
            4'b1011: a_to_g = 7'b0011111;
            4'b1100: a_to_g = 7'b1001111;
            4'b1101: a_to_g = 7'b0111101;

```

```

4'b1110: a_to_g = 7'b1001111;
4'b1111: a_to_g = 7'b1000111;
default: a_to_g = 7'b0000000;

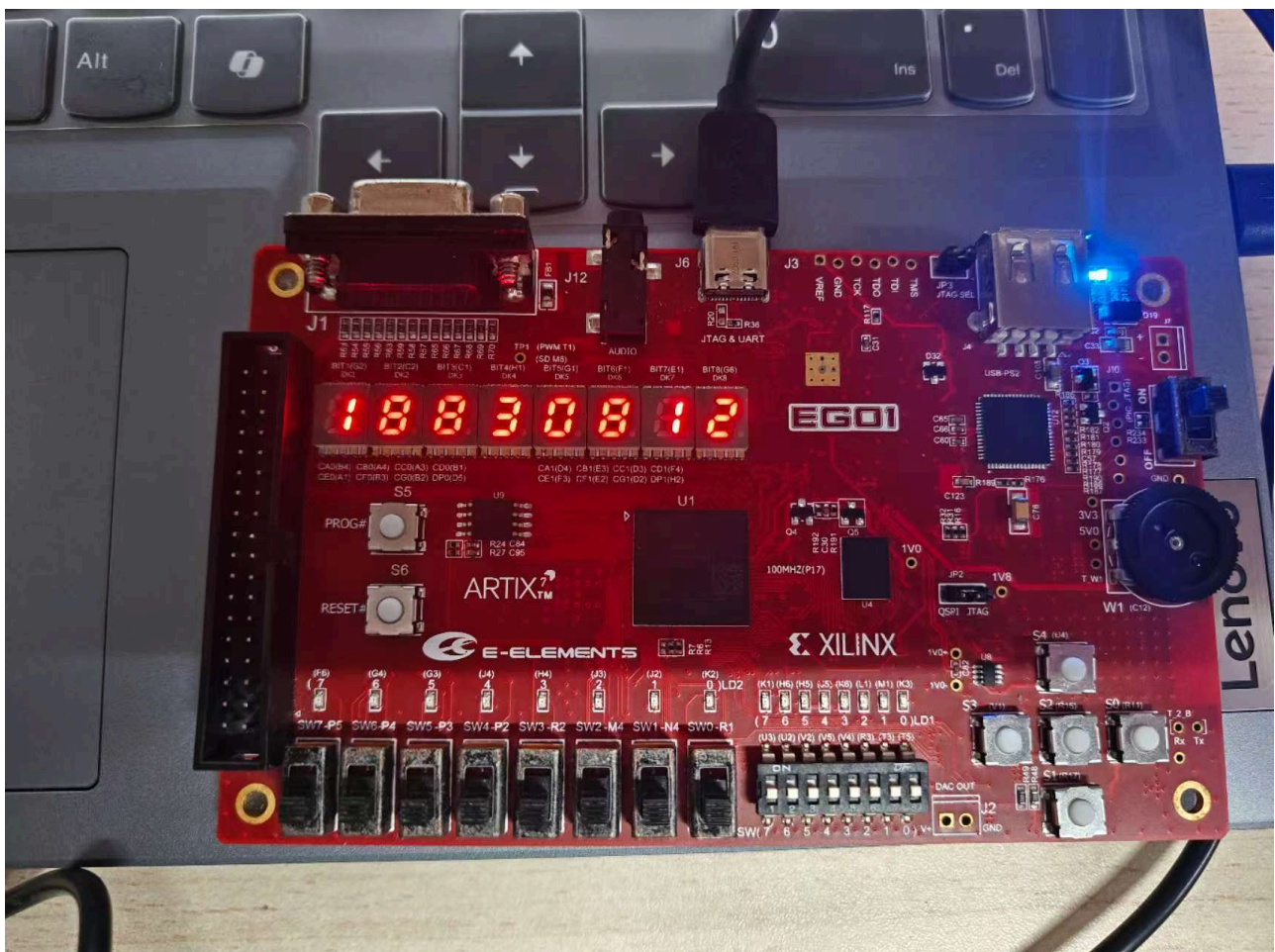
endcase
end

always @ (*) begin // 连续轮流闪亮
    an = 4'b0000;
    an[s] = 1;
end

always @ (posedge clk or posedge reset) begin // 分频器
    if (reset)
        clkdiv <= 0;
    else
        clkdiv <= clkdiv + 1;
end
endmodule

```

上板测试结果

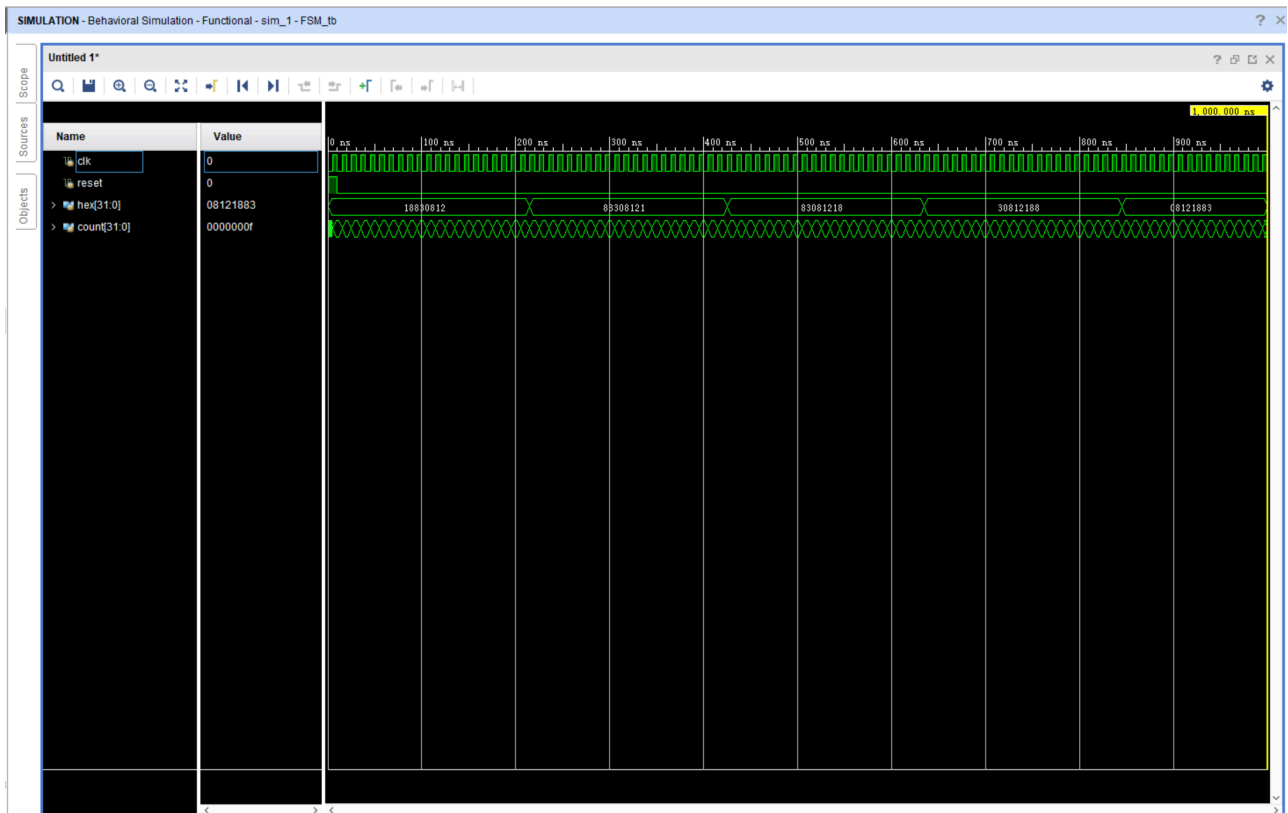


2. 滚起来

代码设计

```
reg [31:0] hex = 32'h18830812;  
  
reg [31:0] count;  
always @(posedge clk) begin  
    if (reset)  
        count <= 0;  
    else if (s) begin  
        if (count == 200000000) begin  
            count <= 0;  
            {hex[31:4], hex[3:0]} <= {hex[27:0], hex[31:28]};  
        end else  
            count <= count + 1;  
        end  
    end  
end
```

仿真测试结果

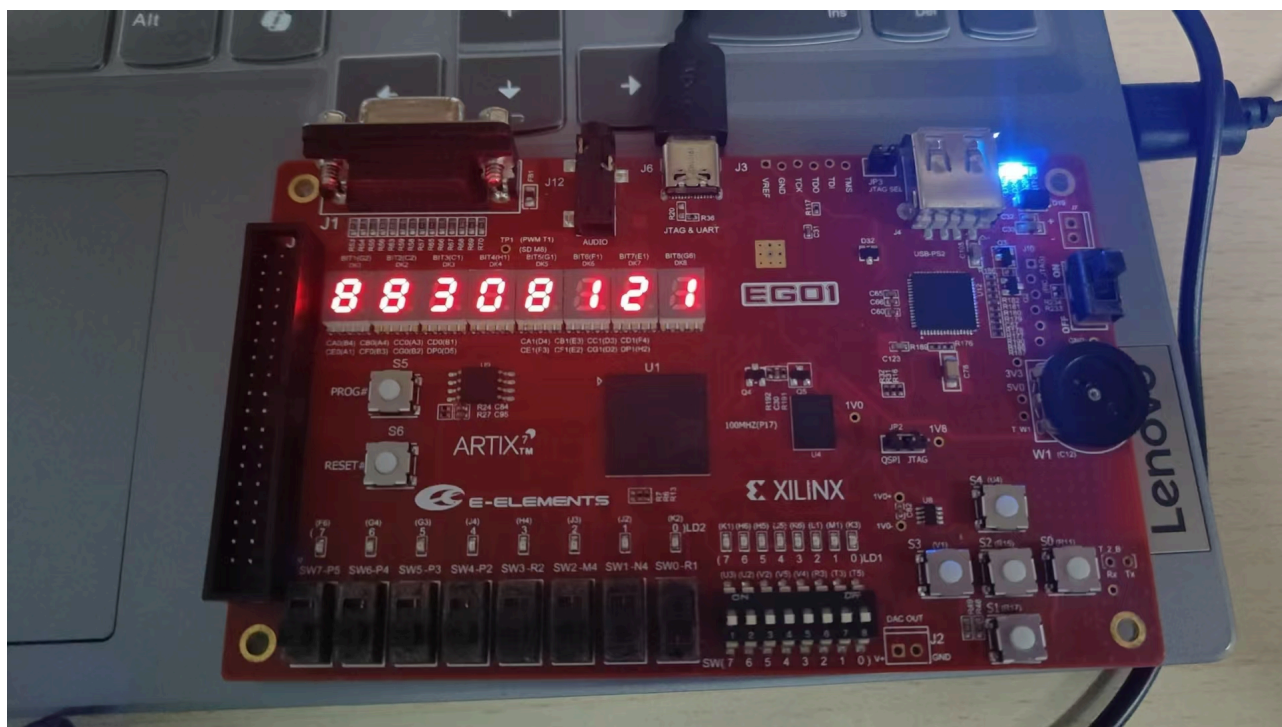


移位寄存器工作正常

3. 顶层代码

```
module top(  
    input s,  
    input clk,  
    input reset,  
    output [6:0] a_to_g0,  
    output [6:0] a_to_g1,  
    output [7:0] an  
);  
    reg [31:0] hex = 32'h18830812;  
  
    reg [31:0] count;  
    always @(posedge clk) begin  
        if (reset)  
            count <= 0;  
        else if (s) begin  
            if (count == 200000000) begin  
                count <= 0;  
                {hex[31:4], hex[3:0]} <= {hex[27:0], hex[31:28]};  
            end else  
                count <= count + 1;  
        end  
    end  
  
    Hex7seg h(.clk(clk), .reset(reset), .hex(hex),  
    .a_to_g0(a_to_g0), .a_to_g1(a_to_g1), .an(an));  
endmodule
```

上板测试结果



实验总结

通过这次实验，我学会了七段数码管的基本使用方法，熟悉了移位寄存器的作用，了解了分频器的应用